Smart City Bike Sharing Data Analysis Using Power BI

1. Project Overview and Objective

This project analyses smart city bike-sharing station data using Power BI. It examines station capacity, available bikes, available docking stands, and occupancy across 25 operating contracts. The report uses interactive visuals and a contract slicer to help users compare station availability and identify locations that may need operational review.

The objective is to build a data-driven dashboard that provides a clear view of the bike-sharing network, compares station availability across contracts, highlights stations with zero available bikes or zero available stands, and supports evidence-based operational investigation.

2. Data Sources

- Source Description and Coverage: Bike-sharing station dataset supplied for the project. The source contains 2,856 station records and 12 original fields, including station details, contract, capacity, available bikes, available stands, status, location coordinates, and last-update timestamp. The Last Update values span 2022–2025. Each station appears once in the fact table; therefore, the data is a station-level snapshot with uneven update dates, not a repeated historical time series.
- Domain: Smart City / Urban Mobility / Bike-sharing Operations Analytics.

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3. Problem Statement

- To describe the bike-sharing network using total stations, bike capacity, available bikes, available stands, occupied stands, and occupancy and availability percentages.
- To compare the number of stations and available bikes across the 25 contracts and identify contracts with relatively high or low availability.
- To identify stations with zero available bikes and stations with zero available docking stands, and compare their counts across contracts.
- To compare occupancy and availability percentages by contract and identify contracts that may require further operational investigation.
- To provide an interactive, station-level view that allows users to filter the dashboard by contract and inspect individual station records.

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4. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
number	Whole Number / Integer	Source station number; combined with Contract Name to create a globally unique StationKey.
name	Text	Name of the bike-sharing station.
address	Text	Street or location address of the station.
position	Text (source)	Original location field containing coordinates; split into Latitude and Longitude in Power Query.
Latitude	Decimal Number	Latitude coordinate created by splitting position.
Longitude	Decimal Number	Longitude coordinate created by splitting position.

banking	Text / Boolean-like	Source indicator describing whether the station supports banking/payment functionality.
bonus	Text / Boolean-like	Source bonus-station indicator.
status	Text	Station status from the source data.
Contract Name	Text	Operating contract or service area associated with the station.
Bike Stands	Whole Number	Total docking capacity of the station.
Available Bike Stands	Whole Number	Number of empty docking spaces currently available.
Available Bikes	Whole Number	Number of bikes available at the station in the source record.
Last Update	Date/Time	Timestamp of the source record's last update.
StationKey	Text	Calculated key: Contract Name combined with station number, used to uniquely identify a station.
Update Date	Date	Date-only value derived from Last Update for the date dimension.
Year	Whole Number	Calendar year derived from Dim_Date[Date].
Month Number	Whole Number	Month number derived from Dim_Date[Date], used to sort Month Name chronologically.
Month Name	Text	Month name derived from Dim_Date[Date].
Quarter	Text	Calendar quarter derived from Dim_Date[Date].

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5. Tools & Technologies

- Power BI Desktop: Data import, Power Query transformations, data modelling, DAX measures, report visuals, and interactive dashboard creation.
- Microsoft Excel: Source-data format and initial data inspection. Power BI Power Query was used for the documented cleaning and transformation steps.

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6. Data Pre-Processing (Power Query)

Tasks Performed:

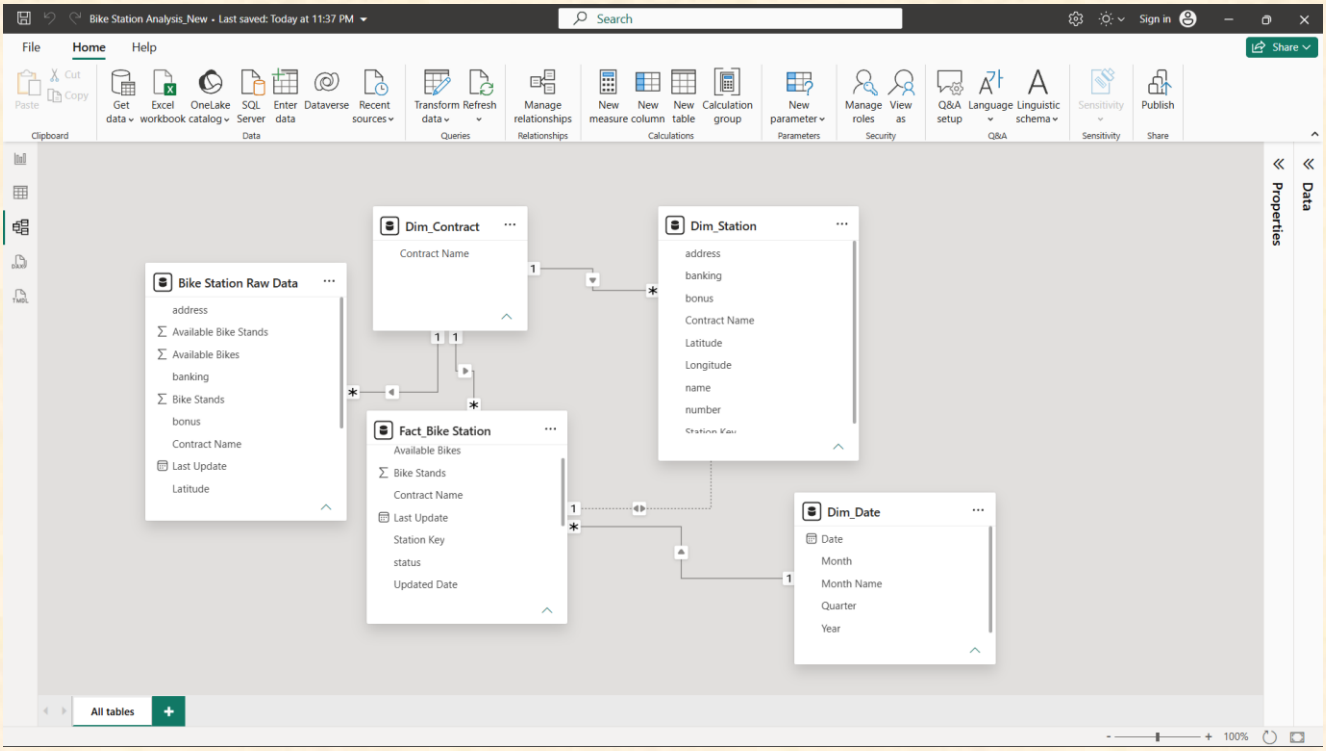
- Data preparation: Imported the source table and checked column data types. Set text, whole-number, date/time, and decimal-number fields to appropriate types. Split the position field into Latitude and Longitude and converted both to decimal numbers.
- Calculated fields: Created StationKey by combining Contract Name and station number. Created Update Date from Last Update for the date dimension. Added Year, Month Number, Month Name, and Quarter attributes to Dim_Date.
- Fact and dimension design: Created Dim_Station, Dim_Contract, Dim_Date, and Fact_BikeStations using Power Query references. Dim_Station contains station attributes; Dim_Contract contains the distinct contract list; Dim_Date contains distinct update dates and calendar attributes; Fact_BikeStations contains station availability and capacity measures.

- Validation: Confirmed 2,856 rows in Fact_BikeStations, 2,856 distinct StationKeys, 25 distinct contracts, and 183 distinct dates in Dim_Date. The documented merge/key checks returned no unmatched records and the prepared tables had no reported errors.

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7. Data Modelling and DAX (Power BI)

- Data Model: Built a star schema with Dim_Station, Dim_Contract, and Dim_Date connected to Fact_BikeStations through active one-to-many relationships. The dimension tables are on the one side and Fact_BikeStations is on the many side.



- DAX Measures: Created the following measures in Fact_BikeStations:
    - Total Available Bikes = SUM(Fact_BikeStations[Available Bikes])
    - Total Bike Capacity = SUM(Fact_BikeStations[Bike Stands])
    - Total Available Stands = SUM(Fact_BikeStations[Available Bike Stands])
    - Total Occupied Stands = [Total Bike Capacity] - [Total Available Stands]
    - Occupancy Percentage = DIVIDE([Total Occupied Stands], [Total Bike Capacity])
    - Availability Percentage = DIVIDE([Total Available Stands], [Total Bike Capacity])
    - Total Stations = DISTINCTCOUNT(Fact_BikeStations[StationKey])
    - Station Occupancy % = DIVIDE(SUM(Fact_BikeStations[Bike Stands]) - SUM(Fact_BikeStations[Available Bike Stands]), SUM(Fact_BikeStations[Bike Stands]))

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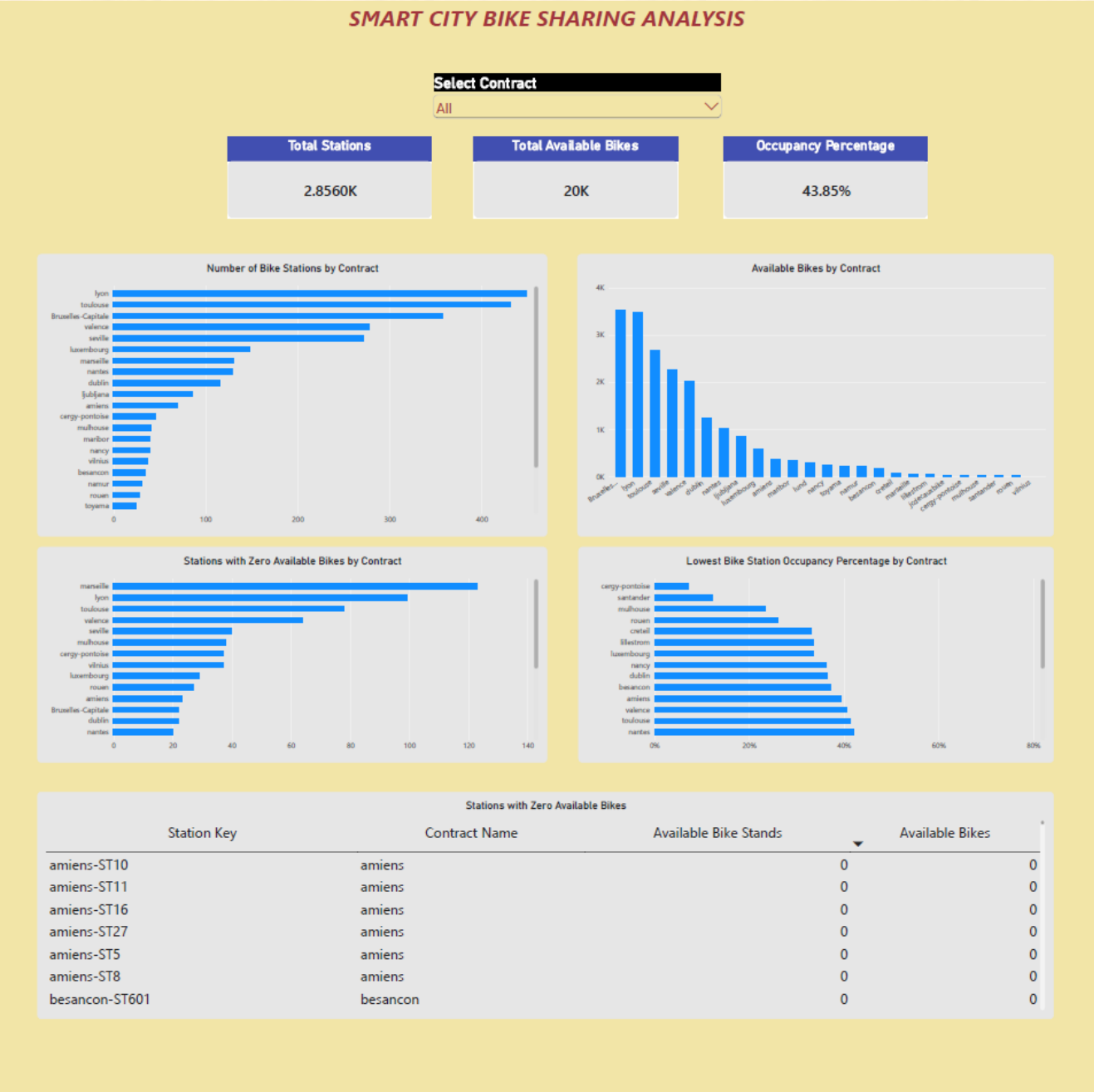
8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Visuals created: KPI cards for Total Available Bikes, Total Bike Capacity, Total Available Stands, Total Occupied Stands, Occupancy Percentage, Availability Percentage, and Total Stations. Charts include Number of Bike Stations by Contract, Available Bikes by Contract, Stations with Zero Available Bikes by Contract, Stations with Zero

Available Stands by Contract, and occupancy/availability percentage by contract. A station-level table displays StationKey, Contract Name, Available Bikes, and Available Bike Stands.

- Interactivity: Added a Contract Name slicer containing all 25 contracts. Tested the slicer with Bruxelles-Capitale, Vilnius, and Marseille; the visuals and station table responded to the selected contract. Clearing the slicer restored the overall values.
- Presentation: Arranged the report on one page with a light background, consistent KPI cards, readable visual titles, and a station-level table across the bottom of the dashboard.
- The consolidated dashboard provides contract-level comparison, network-level KPIs, and station-level detail on one report page.



Stations with Zero Available Bikes

Station Key	Contract Name	Available Bike Stands	Available Bikes
amiens-ST10	amiens	0	0
amiens-ST11	amiens	0	0
amiens-ST16	amiens	0	0
amiens-ST27	amiens	0	0
amiens-ST5	amiens	0	0
amiens-ST8	amiens	0	0
besancon-ST601	besancon	0	0

9. Insights & Conclusions

- Key Findings: The dataset contains 2,856 station records across 25 contracts. Overall occupancy is 43.85% and availability is 56.15%. There are 705 stations with zero available bikes (approximately 24.68% of stations) and 335 stations with zero available stands (approximately 11.73%).
- Four types of analysis:
  - Descriptive: Describes the network using station count, total bike capacity, available bikes, available stands, occupied stands, and overall occupancy/availability percentages.
  - Diagnostic: Highlights patterns that warrant investigation. Vilnius has the highest contract occupancy at 77.92% (availability 22.08%), while Cergy-Pontoise has the lowest occupancy at 7.09% (availability 92.91%). Marseille has the highest absolute count of zero-bike stations (123) and zero-available-stand stations (76). These figures identify areas for investigation; they do not establish the causes.
  - Predictive: The current dataset does not support reliable forecasting of future station availability because each station appears only once and repeated historical availability observations for each station are not available. A future predictive model would require repeated time-stamped station availability or usage data.
  - Prescriptive: Investigate stations with zero available bikes and zero available stands, prioritise contract-level review where counts are high, and consider redistribution only after validating demand, time-of-day patterns, and operational constraints.

10. Conclusions

Power BI enabled the transformation of station-level source data into a structured star-schema model and an interactive dashboard. The report provides a clear descriptive view of bike availability and station occupancy, identifies operational conditions for further investigation, and communicates practical next steps. The uneven Last Update dates and single record per station mean that monthly totals must not be interpreted as a historical network trend or used as a basis for forecasting.